

Project - Low Level Design

On

Multi-Agent Manufacturing System

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Table of Contents

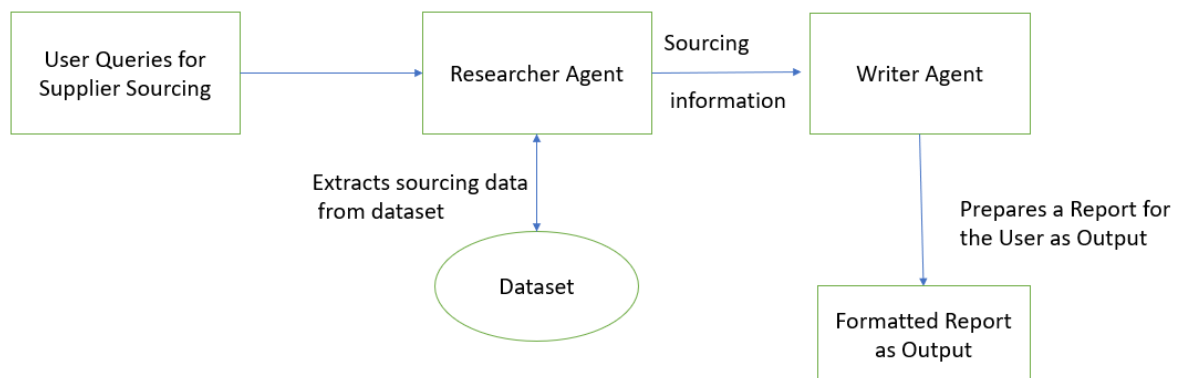
1. Introduction.
 - 1.1. Scope of the document.
 - 1.2. Intended Audience
 - 1.3. System overview.
2. System Design.
 - 2.1. Application Design
 - 2.2. Process Flow.
 - 2.3. Information Flow.
 - 2.4. Components Design
 - 2.5. Key Design Considerations
 - 2.6. API Catalogue.
3. References

1. Introduction

- **1.1. Scope of the Document:** This document provides a detailed low-level design (LLD) for a Multi-Agent Manufacturing System. It covers the technical architecture, specialized agent roles, and the communication protocols required to automate supplier sourcing and report generation.
- **1.2. Intended Audience:** This document is intended for system architects, software developers, and manufacturing stakeholders involved in building or deploying agentic workflows.
- **1.3. System Overview:** The system is a collaborative multi-agent framework designed to handle user queries regarding supplier sourcing. It utilizes specialized agents—a Researcher Agent and a Writer Agent—coordinated by an Orchestrator to transform raw dataset information into structured, high-quality reports.

2. System Design

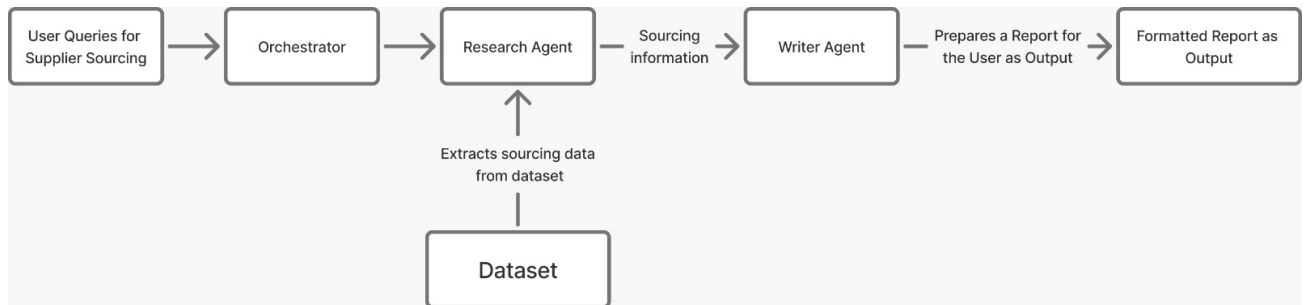
- **2.1. Application Design:** The application is built using a modern AI tech-stack comprising Crew AI for agent orchestration, Langchain for vector embeddings creation, Chroma DB to create a vector database, Streamlit for the user interface, Gemini for high-speed inference, and Python as the core programming language.



- **2.2. Process Flow:**

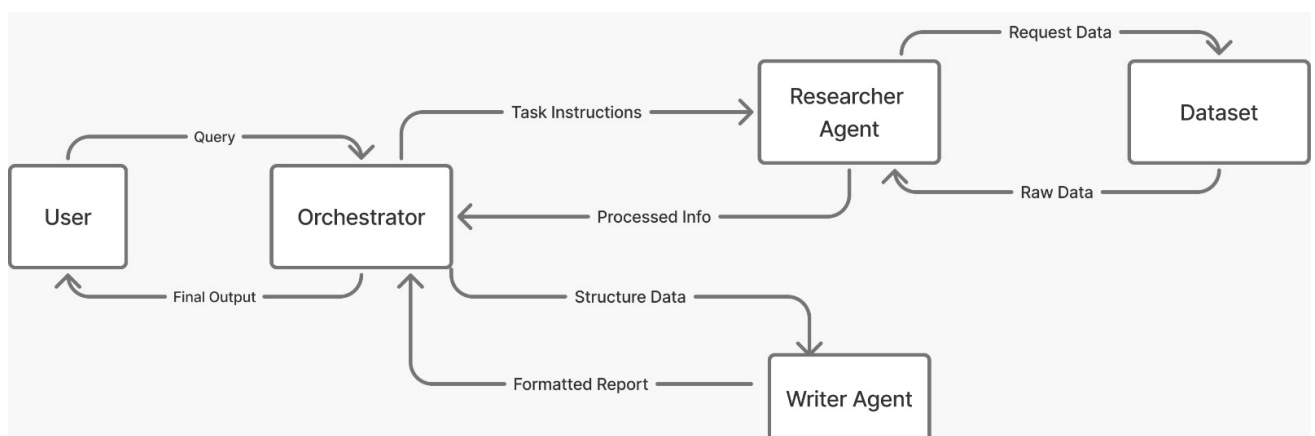
1. **Orchestrator Initiation:** The user provides a prompt, and the Orchestrator initializes the system state.
2. **Researcher Hand-off:** The Orchestrator assigns the task to the Researcher Agent with specific instructions.
3. **Data Gathering:** The Researcher Agent queries the dataset.

4. **Writer Hand-off:** Data is passed back to the Orchestrator and then to the Writer Agent.
5. **Final Delivery:** The Writer Agent creates the response, and the Orchestrator delivers it to the user.



• 2.3. Information Flow:

- **User -> Orchestrator:** Initial query.
- **Orchestrator -> Researcher:** Research parameters.
- **Dataset -> Researcher:** Raw sourcing data.
- **Researcher -> Orchestrator -> Writer:** Sourcing information.
- **Writer -> User:** Formatted report.



• 2.4. Components Design:

- **Researcher Agent:** Specialized in extracting sourcing data from provided datasets.

- **Writer Agent:** Specialized in synthesizing information and preparing structured output for the user.
- **Orchestrator:** Manages state transitions and inter-agent communication protocols.
- **Vector DB:** For storing supplier sourcing information and retrieval of information by researcher agent.
- **2.5. Key Design Considerations:**
 - **Specialization:** Dividing tasks between research and writing to ensure output quality.
 - **State Management:** The Orchestrator maintains the system state throughout the multi-step workflow.
 - **Protocol-Driven Hand-offs:** Clear boundaries and data passing between agents via the Orchestrator.
- **2.6. API Catalogue:**
 - **Crew AI API:** For agent creation and task management.
 - **Gemini API:** For Large Language Model (LLM) interaction and data processing.
 - **Streamlit Components:** For handling user input and displaying the final report.

3. References

- Streamlit Documentation
- Crew AI Documentation
- Langchain Documentation